



Motivation

Problem: Dense embeddings are too costly for large-scale retrieval and deployment.

Gap: Prior efficient methods, especially **CSR** and **MRL**, degrade badly in the **ultra-sparse** regime ($k=2$ or 4).

- Massive Dead Neurons
- Misaligned Self-supervision
- Limited Adaptation Capacity

Prior Methods

- Stable Ultra-Sparse Learning
- Supervision Alignment
- Cross-domain Adaptation

Prior Methods

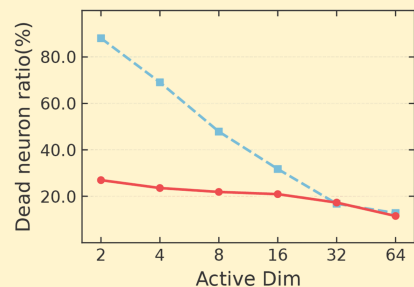
Method

Progressive K-annealing

$$k_t = (1 - p_t) k_{\text{init}} + p_t k_{\text{final}}, p_t = t/T$$

large initial k for exploration $\rightarrow$ small k for target sparsity

— CSR — CSR+anneal

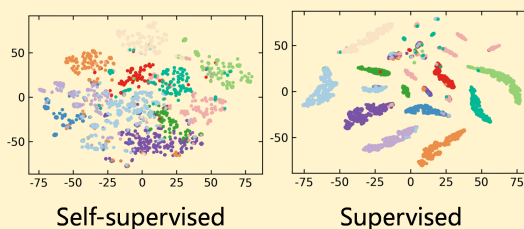


Annealing reduces dead neuron to about 15%.

Sparse Supervised CL

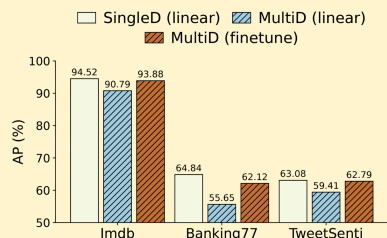
$$\mathcal{L}_{\text{SpCL}} = -\frac{1}{|\mathcal{B}|} \sum_{i=1}^B \log \frac{\exp z_i^T z_i}{\exp z_i^T z_i + \sum_{j \neq i} \exp z_i^T z_j}$$

Sparse Supervised Contrastive Loss facilitates semantic gathering

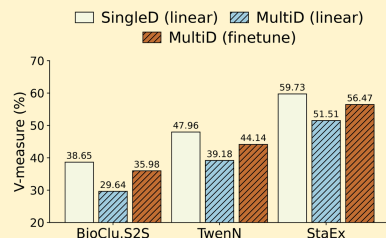


Supervision facilitates feature alignment.

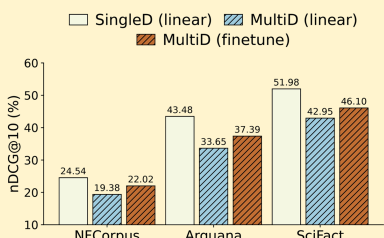
Backbone Finetuning



Classification



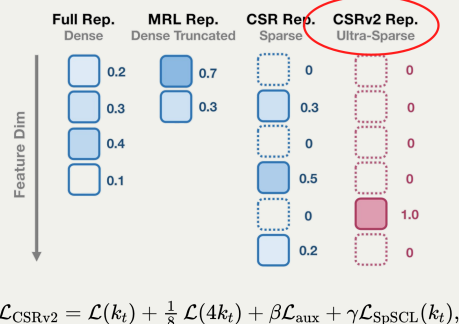
Clustering



Retrieval

Finetuning the full backbone improves robustness across tasks and domains.

Overview



Experiments

More than 300x efficiency gain!

Active Dim	Method	Retrieval Time	Classif. ACC $\uparrow$	Clust. V-measure $\uparrow$	Retrieval nDCG@10 $\uparrow$	STS Spearman $\uparrow$	PairClassif. AP $\uparrow$	Rerank. MAP $\uparrow$	Avg.
4096	eS-Mistral-7B	506.46	80.67	51.55	49.35	84.11	91.77	69.52	69.99
4096	MRL	301.86x	80.46	50.94	48.75	83.78	90.44	68.86	69.49
	CSR	197.52x	80.54	51.13	49.13	83.94	90.99	68.96	69.70
	CSRv2-linear	196.04x	80.55	51.19	49.07	84.02	91.48	69.02	69.76
	CSRv2	201.42x	80.49	51.34	49.16	83.94	91.70	69.18	69.80
64	MRL	17.30x	66.58	47.76	44.11	77.46	78.46	62.72	61.86
	CSR	14.92x	79.50	48.36	45.22	82.10	87.29	64.86	66.68
	CSRv2-linear	14.53x	80.29	48.35	47.92	82.09	88.55	66.54	67.58
	CSRv2	14.17x	79.98	49.53	47.92	82.90	90.46	67.34	68.08
16	MRL	7.77x	54.64	42.03	34.33	68.18	59.22	56.16	51.93
	CSR	3.53x	75.61	45.12	34.79	77.30	84.28	59.86	62.83
	CSRv2-linear	3.51x	77.08	46.58	39.60	79.37	85.38	62.31	64.26
	CSRv2	3.51x	77.79	47.97	43.38	79.94	86.50	64.36	65.76
4	MRL	6.29x	43.84	33.14	24.55	56.51	37.36	44.72	40.83
	CSR	1.62x	67.22	39.25	23.54	70.13	74.44	48.57	52.94
	CSRv2-linear	1.63x	72.55	42.96	34.31	73.31	74.17	56.08	58.62
	CSRv2	1.63x	74.26	43.85	39.04	75.69	74.90	62.93	61.01
2	MRL	6.20x	34.84	26.13	16.63	52.14	26.67	40.30	33.81
	CSR	1.01x	52.50	35.20	16.14	62.93	52.95	46.77	44.33
	CSRv2-linear	1.01x	66.43	39.07	31.58	67.91	57.39	53.32	53.35
	CSRv2	1.00x	71.59	41.29	37.48	73.82	62.46	60.91	58.38

Results with eS-Mistral-7B as backbone

Active Dim	Method	Classif. ACC $\uparrow$	Clust. V-measure $\uparrow$	Retrieval nDCG@10 $\uparrow$	STS Spearman $\uparrow$	PairClassif. AP $\uparrow$	Rerank. MAP $\uparrow$	Avg.
2560	Qwen3-Embed-4B	85.79	55.27	58.37	88.63	91.42	72.03	74.92
2560	MRL	85.38	55.04	58.31	88.02	91.27	71.64	74.58
	CSR	85.49	54.83	58.21	88.64	91.23	71.84	74.70
	CSRv2-linear	85.32	55.43	58.74	89.05	91.03	72.25	74.99
	CSRv2	85.58	55.91	58.23	88.47	91.39	71.98	74.91
64	MRL	83.42	53.73	44.13	86.60	88.08	69.61	70.54
	CSR	83.94	52.36	51.51	85.33	90.54	70.11	71.10
	CSRv2-linear	84.03	53.19	53.22	85.88	90.72	71.13	72.31
	CSRv2	84.28	54.57	55.64	86.32	90.90	71.64	72.79
16	MRL	75.22	47.24	20.40	79.21	73.29	60.82	58.89
	CSR	78.60	49.08	35.66	82.08	85.80	65.00	64.66
	CSRv2-linear	80.71	51.48	39.09	82.15	88.94	67.64	67.20
	CSRv2	82.03	53.86	45.09	82.63	90.42	68.89	68.98
4	MRL	48.27	32.08	6.59	53.11	30.73	40.59	36.74
	CSR	57.39	36.03	16.27	64.13	64.29	50.26	46.66
	CSRv2-linear	71.59	43.24	24.82	72.11	77.74	56.94	56.76
	CSRv2	80.20	48.27	29.71	77.94	82.28	62.98	62.41
2	MRL	26.47	24.20	5.23	30.22	18.46	32.43	22.84
	CSR	51.51	39.22	18.47	51.51	54.60	44.20	36.29
	CSRv2-linear	66.95	39.22	18.47	71.56	77.95	54.67	53.41
	CSRv2	76.22	46.02	23.93	74.88	75.24	59.52	58.53

Comparable Performance with 1/8 neurons!

Results with Qwen3-Embed-4B as backbone

